



Dashboard > Courses > DIP321(English)(1),25/26-P > Topic 11 - Shortest Paths - A*, Bellman-Ford algorithms
> T11 - QUIZ - A* and Bellman-Ford pathfinding algorithms

Algoritmi un programmēšanas metodes(English)(1),25/26-P

T11 - QUIZ - A* and Bellman-Ford pathfinding algorithms

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State Finished

Completed on Friday, 27 March 2026, 1:12 PM

Time taken 24 mins 52 secs

Marks 5.00/5.00

Grade **10.00** out of 10.00 (**100%**)

Question 1

Complete

Mark 1.00 out of 1.00

If your best path found using Bellman Ford algorithm contains a cycle, what does that mean about the graph edges?

Select one:

- ☐ a. There are some 0 weight edges in the graph.
- ☐ b. There are only positive edges in the graph.
- ☐ c. There are only negative edges in the graph.
- ☐ d. There is at least one positive edge in the graph.
- ☒ e. There is at least one negative edge in the graph.



Question 2

Complete

Mark 1.00 out of 1.00

Given $G=\{V,E\}$, how many times do you have to relax all edges (in the worst case) in Bellman Ford algorithm?

PS $|A|$ means number of elements in set A

Select one:

- ☐ a. $|E|+|V|$ times
- ☐ b. $|E| - 1$ times
- ☐ c. $|V|$ times
- ☒ d. $|V|-1$ times
- ☐ e. $|E|$ times

Question 3

Complete

Mark 1.00 out of 1.00

What is admissible heuristic function for A* algorithm?

Select one:

- ☒ a. The heuristic never overestimates the actual cost to get to the goal
- ☐ b. The heuristic always picks the edge with lowest weight.
- ☐ c. The heuristic never picks the lowest cost edge.
- ☐ d. The heuristic always overestimates the actual cost to get to the goal



Question 4

Complete

Mark 1.00 out of 1.00

How does A* differ from Dijkstra algorithm?

Select one:

- ☐ a. None of the mentioned
- ☐ b. Dijkstra supports negative weights, A* does not.
- ☒ c. Dijkstra is like A* algorithm but with heuristic function cost set to 0
- ☐ d. Dijkstra does not support weighted nodes, A* does
- ☐ e. Dijkstra is usually faster than A*

Question 5

Complete

Mark 1.00 out of 1.00

What algorithm can find the optimal path from single source to single destination but does not guarantee shortest paths to other destinations from this single source?

Select one:

- ☐ a. Bellman Ford algorithm
- ☐ b. Dijkstra algorithm
- ☒ c. A* algorithm
- ☐ d. Greedy First search
- ☐ e. Breadth-First-Search

◀ A* search presentations - CMU, Caltech, Lehigh

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Single Source Shortest Path Visualizations ▶

